

Agricultural Statistics

- Agricultural statistics system in India is decentralized, both horizontally and vertically. Primary statistics are collected by the State governments (provincial or sub-national) through **SASA(State Agricultural Statistics Authority)** and consolidated for the country as a whole by the Directorate of Economics & Statistics (DES), under the Department of Agriculture, Cooperation and Farmers' Welfare, under Union MoAFW. The objective of the Directorate(DES) is to provide important statistics on area, production, land use, cost and yield of principal crops, minimum support prices, to implement schemes related to improvement of agricultural statistics.

- **Crop and land use statistics** are basic elements of the Agricultural Statistics System:

1.Crop Area Statistics : From the point of view of crop area statistics, the States and Union Territories can be classified into three broad groups:

1.1 Temporarily settled States (or Land Record States): Statistics of crop area are compiled with the help of the village revenue agency (commonly known as patwari agency).

A complete enumeration of all fields (survey numbers) called **girdawari** is made in every village during each crop season to compile land use, irrigation and crop area statistics.

Due to complete enumeration, girdawari considered fairly reliable and also because of the patwari's intimate knowledge of local agriculture and his ready availability in the village.

An increasing range of functions assigned to the patwari, the girdawari tended to receive low priority.

This system is followed in 18 States and 5 UTs.

1.2 Permanently settled States: No land revenue agency at the village level exists, crop area and land use statistics are collected through a scheme of

sample surveys. Permanently settled states are Kerala, Orissa and West Bengal. Statistics of crop area are compiled by specially appointed field staff under a scheme known as "Establishment of an Agency for Reporting Agricultural Statistics (EARAS). Girdawari is limited to a random sample of 20 per cent villages of the State, which are selected in such a way that during a period of five years, the entire State is covered.

1.3 States with Conventional Estimates: Only —conventional estimates are available and includes Part of Assam (hill districts), Arunachal Pradesh, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim and Tripura, and the two Union Territories of Andaman and Nicobar Islands and Lakshadweep. Tripura and Sikkim (except some minor pockets) are cadastrally surveyed and remaining part is based on conventional crop estimates based on personal assessment of the village chowkidars.

The three categories of States and Union Territories account for eighty-six, nine and five per cent, respectively of the total reporting area.

2. **Land-Use Statistics:** (Nine-fold Classification) Statistics of land use are compiled from the village land records maintained by the patwari. The information is available according to each survey number and recorded under nine categories: (a) Forests, (b) Area under Non-Agricultural use, (c) Barren and Uncultured Land, (d) Permanent Pastures and other Grazing Land, (e) Miscellaneous Tree Crops, (f) Culturable Waste Land, (g) Fallow Land other than Current Fallows, (h) Current Fallows, and (i) Net Area Sown.

- **Crop Production:** Estimates of crop production are obtained by multiplying the area under crop and the yield rate. The yield rate estimates are based on scientifically designed crop cutting experiments conducted under the General Crop Estimation Survey (GCES). The survey design adopted for GCES is that of a stratified three stage random sampling with tehsil or taluka as the stratum, a village as the first stage unit, a field growing the specified crop as the second stage unit and a plot, usually 5m x 5m, as the ultimate unit. The experiment consists of marking the plot and harvesting and weighing the produce from the plot. These weights form the basic data for yield estimation. The number of experiments and their distribution over the strata are made in a manner to be able to obtain the yield rate estimates with a fair degree of precision at the level of the State and each major crop-growing district.

- **Crop Forecasts:** The first forecast relating to the kharif crops is mostly based on reports prepared by the States mainly guided by the visual observation of field officials. The second forecast covering both the kharif and rabi crops takes into account additional information obtained from various sources including agricultural inputs, incidence of pests and diseases, and weekly reports of State departments of agriculture regarding area coverage, conditions of standing crops, etc. Results of Remote Sensing data are also considered at this stage. In the third forecast, the earlier advance estimates of both the kharif and rabi seasons are firmed up, again taking into account information received from sources such as Market Intelligence Units, Meteorological Department and the Crop Weather Watch Group (CWWG). The fourth forecast is based on firm figures supplied by State Agricultural Statistics Authorities (SASAs) who are by then in a position to obtain fairly dependable estimates of yield rates through GCES. In addition to the four forecasts, the DESMOA issues the "Final Estimates" of crop area and production in December. As a few States continue to revise their data on delayed receipt of information, the all-India crop statistics are brought out as "Fully Revised" in the next crop year in the following December.

- Agricultural Statistics (AS) Division, Directorate of Economics & Statistics (DES) releases four Advance Estimates (AE) of production of 28 major agricultural crops followed by Final Estimate of 28 major and 3 minor crops of the country, for every agricultural year (July-June).

Estimate	Released in	Coverage
FAE	September	only Kharif crops
SAE	February	Kharif as well as rabi crops.
TAE	April-May	area coverage for rabi crops and better yield estimates of Kharif crops
FAE	July-Aug	on area as well as yield of Kharif crops and rabi crops
Final Estimate	February (of following year)	area and yield estimates of various crops.

Crop Seasons:

Season	Sowing	Harvested in	Crops
Kharif	July-October(Monsoon)	Sep-Oct	rice, maize, sorghum, pearl millet/bajra, finger millet/ragi (cereals), arhar (pulses), soyabean, groundnut (oilseeds), cotton etc
Rabi	Oct-March(winter)	Mar-Apr	wheat, barley, oats (cereals), chickpea/gram (pulses), linseed, mustard (oilseeds) etc.

- **Agriculture Census**

- It is the prime source of data on structure, distribution and operation of Agriculture Holdings.
- conducted quinquennially** since 1970-71 as part of the World Census of Agriculture (WCA) Programme of the Food and Agriculture Organization (FAO) of the U. N to collect key information about the structure of agriculture holdings in the country and monitor changes that take place over time.
- while World Agriculture Census is conducted decennially.
- The reason for conducting Agriculture Census in India quinquennially is that the structure of holdings in the country is changing very fast due to fragmentation of holdings, urbanization/ industrialization etc.
- Agriculture Year: July-June; Financial Year: April-March; Calendar Year: Jan-Dec
- present Agriculture Census with reference year 2015-16 is 10th ; 11th Agriculture Census with reference year 2020-21 is in progress.
- basic statistical unit for data collection** in Agriculture Census is **“Operational Holding”** rather than **‘Ownership Holding’** as the farm level decisions are taken by persons who operate the land and not by those who own it.
- Central Sector Scheme and is being implemented as a component of **‘Integrated Scheme of Agriculture Census, Economics and Statistics’**.
- Agriculture Census is conducted in the country on census – cum-sample survey basis.
- conducted in **three distinct phases**:
Phase 1- Data on ownership characteristics like gender, social groups and type of ownership is collected using two broad approaches:
(i) In Land-record States, data is collected and compiled through re-tabulation of information available in the Village Land Records (khasra registers). This category covers around 86% of total reported area of the country.
(ii) In Non-Land record States, the data is collected on sample basis following household enquiry method.

Phase 2- data is collected on sample basis from 20 percent villages on detailed characteristics of operational holdings such as tenancy status, land use, irrigation, cropping pattern etc.

Phase 3 (also k/a INPUT SURVEY)- conducted as a follow-up survey to the Agriculture Census (reference year 2016-17 for the 10th Agri. census) to collect data from 7 percent villages on the pattern of input use by operational holdings.

(INPUT SURVEY means information on inputs such as fertiliser, pesticides, agricultural credit, multiple cropping, inventory of agricultural machinery and implements, mechanism of plant protection, quarantine and storage, etc.)

- **Livestock Census**

- by Dept. of Animal Husbandry and Dairying, Min of Fisheries, Animal Husbandry and Dairying

- started in 1919-20; so far 20 censuses have been done (20th: 2019, conducted between oct 2018 to sep 2019)

- conducted quinquennially; a complete count of the livestock and poultry at pre-defined reference point of time.

- Livestock Census is the only source of statistics on livestock numbers, their age and sex structure, and functional classification

- Why Agriculture Census and Livestock Census can not be conducted simultaneously?

(a) The basic unit of enumeration in the Agricultural Census is an "operational holding" whereas in the Livestock Census it is a "household". (b) The reference period for the Agricultural Census is one year whereas it is a specific date for the count of numbers in the Livestock Census.

(c) Agricultural Census is conducted through a dual programme of census and sample survey whereas the Livestock Census is based on cent per cent coverage of all households in the country (urban and rural).

(d) Different field agencies are used for the two censuses, the patwari agency for the Agricultural Census and the field staff of the State Department of Animal Husbandry for the Livestock Census.

However, these problems can be overcome as;

(a) Like the Livestock Census, the Agricultural Census also follows the household approach in the permanently settled States where it has been possible to collect information on operational holdings through household enquiry. Even in the temporarily settled States, where the Agricultural Census is based on the re-tabulation of land records, there is an element of household enquiry to account for the details of land held by resident holders outside the village precincts.

(b) As regards different reference periods of the two censuses, the Livestock Census has never been able to adhere to the stipulation of a specific date for the count of numbers. On the contrary, the census is spread over several years. If the Agricultural and Livestock Censuses are synchronised to be taken during the same year, the livestock count could be concentrated around the specified reference date as far as possible with appropriate check of the change in numbers between the date of enquiry and the reference date.

(c) Both the censuses be limited to a 20 per cent sample of villages, which also facilitates the process of integration.

(d) There is a definite advantage in entrusting the field operations to a single reporting agency with enough safeguards such as careful advance planning and by ensuring that the censuses are accorded the requisite priority by the State administration as in the case of the Population Census.

- In order to improve the timeliness and quality of crop area statistics, A scheme —**Improvement of Agricultural Statistics(IAS)** is in operation since early seventies. It has three major components: Timely Reporting Scheme (TRS), Establishment of an Agency for Reporting of Agricultural Statistics (EARAS), Improvement of Crop Statistics (ICS).
1. **Timely Reporting Scheme (TRS):** TRS is being implemented in 16 land record States. The objective of this component is to obtain estimates of area under principal crops in each season, with the breakup of area under irrigated/unirrigated and traditional /high yielding varieties of the crops on the basis of random sample of 20% of villages by a specific date. These estimates are used for generating advance estimates of production of principal crops. The objective of the Scheme is also to reduce time lag in making available the area statistics of major crops in addition to providing the sampling frame for selection of crop-growing fields for crop cutting experiments. Under the TRS, the patwari is required to complete the girdawari on a priority basis in a 20 per cent random sample of villages and to submit the village crop statements to higher authorities by a stipulated date for the preparation of advance estimates of the area under major crops. These are used in the framing of crop forecasts. The TRS sample of villages is also selected in such a way that the entire temporarily settled parts of the country are covered over a period of five years.
 2. **Establishment of an Agency for Reporting of Agricultural Statistics (EARAS):** EARAS is implemented in three non-land record States, namely, Kerala, Orissa and West Bengal and four North Eastern States, namely, Arunachal Pradesh, Nagaland, Sikkim and Tripura. In the absence of any Government Department for the regular updating of land records in these States, a full time agency for collection of statistics of area, production and yield rate has been established in each of the implementing States. The basic methodology under this component is the same as in TRS.

3. **Improvement of Crop Statistics (ICS):** Under the ICS scheme, an independent agency of supervisors carries out a physical verification of the patwari's girdawari in a sub-sample of the TRS sample villages (in four clusters of five survey numbers each); and makes an assessment of the extent of discrepancies between the supervisor's and patwari's crop area entries in the sample clusters. The supervisor also scrutinises the village crop abstract prepared by the patwari and checks whether it is free from totaling errors and whether it has been dispatched to the higher authorities by the stipulated time.

- **Comprehensive Scheme for Studying Cost of Cultivation of Principal Crops in India**

- initiated in 1970-71

- Directorate of Economics & Statistics (DES), under the Department of Agriculture, Cooperation and Farmers' Welfare, under Union MoAFW has the overall charge of implementing the survey programme through the Agricultural Universities and other institutes.

- The Cost of cultivation/ Production Estimates being generated with the data collected under the scheme are used as an important input by CACP while recommending MSP of various crops. Apart from this, the data is also used by MoSPI for preparation of National Accounts Statistics. Marketed Surplus Ratio which is required for various policy decisions is also compiled from the data.

- covers major 25 crops

- Method:

Each state is divided into homogenous agro-climatic zones based on cropping pattern, soil types, rainfall etc. Each zone is a strata. From each strata three-stage sampling is adopted.

Ist Stage: Selection of Tehsil/Block by PPSWR with size measure as 'Area under the crop complex'. So the f.s.u. are 'tehsils'.

IIInd Stage: Selection of village/cluster by PPSWR with size measure as 'Area under the crop complex'. So the s.s.u. are village/cluster of villages.

IIIrd Stage: Selection of 'Operational Holding' in each selected village by SRSWOR from each of 5 land-size classes. In each size class, two holdings are selected by SRSWOR. So the t.s.u is Operational Holdings.

Classification of operational holdings or land-size class- (i) Less than 1 hectare (ii) Between 1-2 hectare (iii) Between 2-4 hectare (iv) Between 4-6 hectare (v) Above 6 hectare

The fieldwork consists of collecting from each sample household through the cost accounting method, data on all aspects of cultivation (inputs, outputs, prices paid and received) by keeping a detailed record on a day-to-day basis.

- **Forecasting Agricultural output using Space, Agro-meteorology and Land based observations (FASAL)**

Component-wise details of the Scheme: The scheme has the following partner agencies:-

- India Meteorological Department, New Delhi
- Institute of Economic growth, New Delhi
- Space Application Centre, Ahmadabad
- National Remote Sensing Centre, Hyderabad

India Meteorological Department India Meteorological Department (IMD) has been developing crop yield forecasting models based on multiple correlation and regression techniques in collaboration with Agromet Field Units (AMFUs) using both Statistical and Crop Growth Stimulation Models. Crop yield forecast of major crops is issued at mid-season and pre-harvest stage using agromet models during Kharif as well as Rabi season for crops. Presently, IMD has been providing crop yield forecast in respect of Rice, Jute, Cotton and Sugarcane during Kharif season and Wheat, Potato, Mustard and Sorghum during Rabi season to MNCFC for the preparation of forecasts based on Remote Sensing.

Institute of Economic Growth (IEG) IEG provides State level / National level forecast of area, yield and production forecasts in Kharif and Rabi Season at pre-sowing (F0) and sowing (F1) stage based on Econometric models. Main functions/objectives of IEG related to FASAL are as follows:-

- To develop econometric models for providing forecasts at National and State level for major crops.
- Using Econometric Analysis to make crop-forecast at pre-sowing stage and Plantation stage.
- Providing forecasts of area and production of selected crops based on economic parameters much in advance of the estimated generated by conventional system.

So far, IEG was providing forecast for 12 crops at pre-sowing(F0) and sowing stage(F1) for Kharif season and for 9 crops at pre-sowing(F0) and sowing(F1) stage for rabi season selected stages. IEG has now been mandated to give monthly forecast of Area and Production of 13 major crops under RTISA. IEG has started submitting monthly forecast to CFCC Division of DES.

Space Application Centre (SAC), Ahmedabad SAC, Ahmedabad develops Remote Sensing Methodology for 11 Crops. SAC provides forecast at the mid season & pre-harvest stage of crop growth when crop is visible on the ground and crop coverage is amenable to RS techniques. So far, Remote Sensing (RS) methodologies in respect of 8 crops i.e., Wheat, Rice (Kharif), Potato (winter), Rapeseed & Mustard, Jute, Rabi Sorghum, Sugarcane and Cotton have been transferred to Mahalanobis National Crop Forecasting Centre (MNCFC). The methodologies in respect of rest of the crops i.e. Kharif Sorghum, Kharif Maize, Pearl Millet, Finger Millet and Groundnut are being under development.

National Remote Sensing Centre (NRSC) NRSC has been providing satellite data to MNCFC for operationalizing the remote sensing methodologies developed by ISRO. The main function of NRSC data Centre is to purchase various satellite data for different crops during Kharif and Rabi season for MNCFC in order to operationalize the RS methodologies developed and transferred by SAC Ahmadabad to MNCFC. All the operational components of FASAL and NADAMS project are, therefore, being transferred from SAC, Ahmedabad and NRSC, Hyderabad to the MNCFC. Further refinement & development will continue to take place in SAC and NRSC.

- **Small Area Estimation Techniques:** Small area typically refers to the part of a population for which reliable statistics of interest cannot be produced due to small sample sizes. Among the implicitly model based methods, the synthetic method of estimation has been used in variety of situations. It is as follows- : An unbiased estimate is obtained from a sample survey for a larger area. When this estimate is used to derive estimates for sub-areas having the same characteristics as the larger area, these estimates are identified as synthetic estimates. The approach is also called —scaling down|| approach. Use of auxiliary information at the basic level is crucial for scaling down the larger area estimates to smaller area. The synthetic approach of estimation is based on the assumption that the small area behaves similarly as the large area. When this does not hold the synthetic estimate may turn out to be heavily biased and thus may not give a satisfactory estimate.

- INDEX NUMBERS IN AGRICULTURE by DES, MoAFW
 - (a) Index Numbers of Area, Production and Yield
 - (b) Index Numbers of Terms of Trade between Agricultural and Non-agricultural Sectors.

- QuickShot:

1. State Agricultural Statistics Authority (SASA) is associated with State Agriculture. State Governments have designated one of its Departments as SASA which is responsible for providing requisite data at prescribed intervals to DES for finalizing Advance/Final Estimates. SASA's vary from State to State and is one of the following State Departments:
 - Deptt. of Agriculture (Gujarat, Haryana, Jharkhand, Maharashtra Punjab, UP, Uttarakhand and WB)
 - Deptt. of Economics & Statistics (AP, Assam, Bihar, J&K, Karnataka, Kerala, Orissa, Rajasthan and TamilNadu)
 - Deptt. of Land Records (Chattisgarh, HP and MP)

2. Directorate of Marketing Intelligence(DMI) under Ministry of Agriculture is primarily responsible to collect price data on Agricultural products.
3. Commission for Agricultural Costs and Prices (CACP) recommends MSP (Minimum Support Price) for crops.
4. Khasra Register is Village Crop Register.
5. An operational holding is defined as "all land, which is used wholly or partly for agricultural production and is operated as one technical unit by one person alone or with others without regard to title, legal form, size or location"
6. Forest Survey of India (FSI) was created in 1981 with the objective of monitoring the forest resources at a macro level, storing and retrieving forestry related data, designing methodology for forest surveys, etc.
7. Since 1987, the FSI has begun using Remote Sensing (RS) technology to collect data on forest cover under three broad classes (dense forest, open forest and mangroves) on a country-wide scale through a biennial survey.
8. Crop Statistics has 2 components- Area sown and Average yield
9. DAC&FW has established Mahalanobis National Crop Forecast Centre (MNCFC) on April 23, 2012 which operationalises Remote Sensing Methodology on crop forecast and drought assessments in collaboration with ISRO. These forecasts are used to validate the official crop estimates based the conventional system.